Extrema and the Mean Value Theorem

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Critical points and extrema

Find the critical points of $f(x) = 2x^3 - 3x^2 - 12x + 5$.

$f'(x) = 6x^2 - 6x - 12 = 6(x^2 - x - 2) = 6(x - 2)(x + 1)$. Critical points: $x = 2$ and $x = -1$.

2 [Foundational] • Target: Critical points and extrema

Find the absolute maximum and minimum of $f(x) = x^3 - 3x$ on $[-2, 3]$, checking critical points and endpoints.

$f'(x) = 3x^2 - 3 = 0$ gives $x = \pm 1$ (both in $[-2, 3]$). Values: $f(-2) = -2$, $f(-1) = 2$, $f(1) = -2$, $f(3) = 18$. Absolute max: $f(3) = 18$. Absolute min: $f(-2) = f(1) = -2$ (tied).

3 [Foundational] • Target: The Extreme Value Theorem

State the Extreme Value Theorem and explain why it guarantees $f(x) = x^2 - 4x$ has an absolute max and min on $[0, 5]$.

The EVT states that a continuous function on a closed interval $[a, b]$ attains both an absolute maximum and an absolute minimum on that interval. Since $f(x) = x^2 - 4x$ is continuous (a polynomial) and $[0, 5]$ is closed, the EVT guarantees both extrema exist.

4 [Foundational] • Target: Rolle's Theorem

Verify that Rolle's Theorem applies to $f(x) = x^2 - 4x + 3$ on $[1, 3]$, and find the value(s) of c guaranteed by the theorem.

$f(1) = 0$ and $f(3) = 0$: equal endpoint values, and f is continuous and differentiable everywhere, so Rolle's Theorem applies. $f'(x) = 2x - 4 = 0$ gives $c = 2$, which lies in $(1, 3)$.

5 [Foundational] • Target: Rolle's Theorem

Explain why Rolle's Theorem does NOT apply to $f(x) = |x - 2|$ on $[0, 4]$, even though $f(0) = f(4) = 2$.

f is not differentiable at $x = 2$ (it has a corner), so one of Rolle's Theorem's hypotheses (differentiability on the open interval) fails — the theorem's guarantee does not apply.

6 [Foundational] • Target: The Mean Value Theorem

For $f(x) = x^3$ on $[0, 2]$, find the value(s) of c guaranteed by the Mean Value Theorem.

Average rate of change: $\frac{f(2) - f(0)}{2 - 0} = \frac{8}{2} = 4$. Need $f'(c) = 3c^2 = 4$: $c^2 = \frac{4}{3}$, $c = \frac{2}{\sqrt{3}} \approx 1.15$ (taking the positive root within $[0, 2]$).

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7 [Foundational] • Target: The Mean Value Theorem

A car travels 180 km in 2 hours. Using the Mean Value Theorem, explain why the car must have been traveling at exactly 90 km/h at some instant.

Average velocity $= \frac{180}{2} = 90$ km/h. Since position is a differentiable function of time, the MVT guarantees some instant c where the instantaneous velocity $v(c) = 90$ km/h exactly.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Using the MVT to bound function values

Given $f(2) = 5$ and $-3 \le f'(x) \le 4$ for all x, use the MVT to find the range of possible values for $f(6)$.

By the MVT, $f(6) - f(2) = f'(c)(6-2) = 4f'(c)$ for some $c \in (2, 6)$. Since $-3 \le f'(c) \le 4$: $-12 \le f(6) - 5 \le 16$, so $-7 \le f(6) \le 21$.

9 [Intermediate] • Target: Consequences of the MVT: monotonicity

Explain, using the MVT, why $f'(x) > 0$ on an interval implies f is strictly increasing on that interval.

For any $a < b$ in the interval, the MVT gives $\frac{f(b) - f(a)}{b - a} = f'(c)$ for some $c \in (a, b)$. Since $f'(c) > 0$ and $b - a > 0$, we get $f(b) - f(a) > 0$, i.e. $f(b) > f(a)$ — so f is strictly increasing.

10 [Intermediate] • Target: Consequences of the MVT: monotonicity

Explain why two functions f and g with $f'(x) = g'(x)$ for all x must differ by a constant, using the MVT.

Let $h(x) = f(x) - g(x)$. Then $h'(x) = f'(x) - g'(x) = 0$ for all x. If h were not constant, the MVT applied between any two points where h differs would give $h'(c) \neq 0$ for some c, a contradiction. So $h(x) = C$ for some constant C, i.e. $f(x) = g(x) + C$.

11 [Intermediate] • Target: Visualizing the Mean Value Theorem

The graph shows $f(x) = x^3$ on $[0, 2]$ together with the secant line through the endpoints and the tangent line at $x = \frac{2}{\sqrt{3}}$ (found earlier).

The tangent line at $c = \frac{2}{\sqrt{3}}$ is parallel to the secant line connecting $(0, 0)$ and $(2, 8)$, both having slope 4 — this is exactly the geometric statement of the MVT: somewhere between the endpoints, the instantaneous slope matches the average slope.

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12 [Intermediate] • Target: Absolute extrema on a closed interval — a capstone problem

Find the absolute maximum and minimum of $f(x) = x^4 - 4x^2 + 3$ on $[-2, 2]$, checking all critical points and endpoints.

$f'(x) = 4x^3 - 8x = 4x(x^2 - 2) = 0$ gives $x = 0, \pm\sqrt{2}$, all within $[-2, 2]$. Values: $f(-2) = 16 - 16 + 3 = 3$, $f(-\sqrt{2}) = 4 - 8 + 3 = -1$, $f(0) = 3$, $f(\sqrt{2}) = -1$, $f(2) = 3$. Absolute max: 3 (at $x = -2, 0, 2$). Absolute min: -1 (at $x = \pm\sqrt{2}$).

13 [Intermediate] • Target: Rolle's Theorem with trigonometric functions

Verify that Rolle's Theorem applies to $f(x) = \sin x$ on $[0, \pi]$, and find the value of c guaranteed by the theorem.

$f(0) = 0$ and $f(\pi) = 0$: equal endpoint values, and $\sin x$ is continuous and differentiable everywhere, so Rolle's Theorem applies. $f'(x) = \cos x = 0$ gives $c = \frac{\pi}{2}$, which lies in $(0, \pi)$.

14 [Intermediate] • Target: An applied Mean Value Theorem problem

A runner completes a 10 km race in 40 minutes. Using the Mean Value Theorem, find the runner's average speed, and explain why there must be some instant during the race when the runner's instantaneous speed equals this average.

Average speed $= \frac{10}{40} = 0.25$ km/min $= 15$ km/h (distance divided by time). Since distance is a differentiable function of time, the MVT guarantees some instant c during the race where the instantaneous speed equals exactly 15 km/h.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The Extreme Value Theorem • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Given $f(1) = -5$ and $-2 \le f'(x) \le 2$ for all x, use the MVT to find the largest possible value of $f(2)$.

Final Answer: -3. By the MVT, $f(2) - f(1) = f'(c)(1)$ for some $c \in (1, 2)$. The largest value occurs when $f'(c) = 2$: $f(2) = -5 + 2(1) = -3$.

16 [Challenging] • Target: Rolle's Theorem • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Given $f(-3) = 3$ and $-3 \le f'(x) \le 5$ for all x, use the MVT to find the largest possible value of $f(-1)$.

Final Answer: 13. By the MVT, $f(-1) - f(-3) = f'(c)(2)$ for some $c \in (-3, -1)$. The largest value occurs when $f'(c) = 5$: $f(-1) = 3 + 5(2) = 13$.

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17 [Challenging] • Target: The Mean Value Theorem • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Given $f(-4) = -3$ and $-6 \le f'(x) \le 2$ for all x, use the MVT to find the largest possible value of $f(2)$.

Final Answer: 9. By the MVT, $f(2) - f(-4) = f'(c)(6)$ for some $c \in (-4, 2)$. The largest value occurs when $f'(c) = 2$: $f(2) = -3 + 2(6) = 9$.

18 [Challenging] • Target: Using the MVT to bound function values • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Given $f(1) = 1$ and $-2 \le f'(x) \le 5$ for all x, use the MVT to find the largest possible value of $f(4)$.

Final Answer: 16. By the MVT, $f(4) - f(1) = f'(c)(3)$ for some $c \in (1, 4)$. The largest value occurs when $f'(c) = 5$: $f(4) = 1 + 5(3) = 16$.

19 [Challenging] • Target: Consequences of the MVT: monotonicity • Cross-Lesson: 8.01 Area Between Curves

Given $f(3) = 5$ and $-2 \le f'(x) \le 3$ for all x, use the MVT to find the largest possible value of $f(6)$.

Final Answer: 14. By the MVT, $f(6) - f(3) = f'(c)(3)$ for some $c \in (3, 6)$. The largest value occurs when $f'(c) = 3$: $f(6) = 5 + 3(3) = 14$.

20 [Challenging] • Target: Visualizing the Mean Value Theorem • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Find the larger critical point of $f(x) = x^3 - 9x^2 + 24x$.

Final Answer: 4. $f'(x) = 3x^2 - 18x + 24 = 3(x - (2))(x - (4))$. Critical points: $x = 2$ and $x = 4$; the larger is 4.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Critical points and extrema

Verify that Rolle's Theorem applies to $f(x) = x^2 + 4x$ on $[-6, 2]$, and find the value of c guaranteed by the theorem.

a. -2 ✓

b. -6

c. -1

d. 2

$f(-6) = f(2)$ (both equal by symmetry), and f is a polynomial (continuous and differentiable everywhere), so Rolle's Theorem applies. $f'(x) = 2x + 4 = 0$ gives $c = -2$, which lies in $(-6, 2)$.

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22 [Foundational] • Target: The Extreme Value Theorem

Verify that Rolle's Theorem applies to $f(x) = x^2$ on $[-2, 2]$, and find the value of c guaranteed by the theorem.

a. 1

b. 0 ✓

c. 2

d. -2

$f(-2) = f(2)$ (both equal by symmetry), and f is a polynomial (continuous and differentiable everywhere), so Rolle's Theorem applies. $f'(x) = 2x = 0$ gives $c = 0$, which lies in $(-2, 2)$.

23 [Foundational] • Target: Rolle's Theorem

Verify that Rolle's Theorem applies to $f(x) = x^2 - 10x$ on $[0, 10]$, and find the value of c guaranteed by the theorem.

a. 5 ✓

b. 10

c. 6

d. 0

$f(0) = f(10)$ (both equal by symmetry), and f is a polynomial (continuous and differentiable everywhere), so Rolle's Theorem applies. $f'(x) = 2x - 10 = 0$ gives $c = 5$, which lies in $(0, 10)$.

24 [Foundational] • Target: The Mean Value Theorem

Verify that Rolle's Theorem applies to $f(x) = x^2 + 4x$ on $[-5, 1]$, and find the value of c guaranteed by the theorem.

a. -1

b. -2 ✓

c. -5

d. 1

$f(-5) = f(1)$ (both equal by symmetry), and f is a polynomial (continuous and differentiable everywhere), so Rolle's Theorem applies. $f'(x) = 2x + 4 = 0$ gives $c = -2$, which lies in $(-5, 1)$.

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25 [Foundational] • Target: Using the MVT to bound function values

Verify that Rolle's Theorem applies to $f(x) = x^2 + 12x$ on $[-9, -3]$, and find the value of c guaranteed by the theorem.

a. -5

b. -6 ✓

c. -3

d. -9

$f(-9) = f(-3)$ (both equal by symmetry), and f is a polynomial (continuous and differentiable everywhere), so Rolle's Theorem applies. $f'(x) = 2x + 12 = 0$ gives $c = -6$, which lies in $(-9, -3)$.

26 [Foundational] • Target: Consequences of the MVT: monotonicity

Verify that Rolle's Theorem applies to $f(x) = x^2 + 10x$ on $[-6, -4]$, and find the value of c guaranteed by the theorem.

a. -3

b. -6

c. -4

d. -5 ✓

$f(-6) = f(-4)$ (both equal by symmetry), and f is a polynomial (continuous and differentiable everywhere), so Rolle's Theorem applies. $f'(x) = 2x + 10 = 0$ gives $c = -5$, which lies in $(-6, -4)$.

27 [Foundational] • Target: Visualizing the Mean Value Theorem

Verify that Rolle's Theorem applies to $f(x) = x^2 - 6x$ on $[2, 4]$, and find the value of c guaranteed by the theorem.

a. 4

b. 5

c. 2

d. 3 ✓

$f(2) = f(4)$ (both equal by symmetry), and f is a polynomial (continuous and differentiable everywhere), so Rolle's Theorem applies. $f'(x) = 2x - 6 = 0$ gives $c = 3$, which lies in $(2, 4)$.

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28 [Foundational] • Target: Absolute extrema on a closed interval — a capstone problem

Verify that Rolle's Theorem applies to $f(x) = x^2 - 12x$ on $[1, 11]$, and find the value of c guaranteed by the theorem.

a. 6 ✓

b. 1

c. 7

d. 11

$f(1) = f(11)$ (both equal by symmetry), and f is a polynomial (continuous and differentiable everywhere), so Rolle's Theorem applies. $f'(x) = 2x - 12 = 0$ gives $c = 6$, which lies in $(1, 11)$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Rolle's Theorem with trigonometric functions

Given $f(1) = -5$ and $-4 \le f'(x) \le 4$ for all x, use the MVT to find the largest possible value of $f(7)$.

a. -29

b. 25

c. 15

d. 19 ✓

By the MVT, $f(7) - f(1) = f'(c)(6)$ for some $c \in (1, 7)$. The largest value occurs when $f'(c) = 4$: $f(7) = -5 + 4(6) = 19$.

30 [Intermediate] • Target: An applied Mean Value Theorem problem

Given $f(-3) = 6$ and $-4 \le f'(x) \le 5$ for all x, use the MVT to find the largest possible value of $f(5)$.

a. 41

b. 46 ✓

c. -26

d. 54

By the MVT, $f(5) - f(-3) = f'(c)(8)$ for some $c \in (-3, 5)$. The largest value occurs when $f'(c) = 5$: $f(5) = 6 + 5(8) = 46$.

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31 [Intermediate] • Target: Critical Points & First Derivative Test

Given $f(2) = 1$ and $-5 \le f'(x) \le 4$ for all x, use the MVT to find the largest possible value of $f(9)$.

a. -34

b. 36

c. 25

d. 29 ✓

By the MVT, $f(9) - f(2) = f'(c)(7)$ for some $c \in (2, 9)$. The largest value occurs when $f'(c) = 4$: $f(9) = 1 + 4(7) = 29$.

32 [Intermediate] • Target: Mean Value Theorem (MVT)

Given $f(1) = -7$ and $-3 \le f'(x) \le 6$ for all x, use the MVT to find the largest possible value of $f(9)$.

a. 41 ✓

b. 35

c. 49

d. -31

By the MVT, $f(9) - f(1) = f'(c)(8)$ for some $c \in (1, 9)$. The largest value occurs when $f'(c) = 6$: $f(9) = -7 + 6(8) = 41$.

33 [Intermediate] • Target: Absolute Extrema on Closed Intervals (Candidates Test)

Given $f(4) = 8$ and $-3 \le f'(x) \le 4$ for all x, use the MVT to find the largest possible value of $f(6)$.

a. 12

b. 18

c. 16 ✓

d. 2

By the MVT, $f(6) - f(4) = f'(c)(2)$ for some $c \in (4, 6)$. The largest value occurs when $f'(c) = 4$: $f(6) = 8 + 4(2) = 16$.

34 [Intermediate] • Target: Critical points and extrema

Find the larger critical point of $f(x) = x^3 + 6x^2 - 36x$.

a. -4

b. 2 ✓

c. -6

d. -12

$f'(x) = 3x^2 + 12x - 36 = 3(x - (-6))(x - (2))$. Critical points: $x = -6$ and $x = 2$; the larger is 2.

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35 [Intermediate] • Target: The Extreme Value Theorem

Find the larger critical point of $f(x) = x^3 + 9x^2$.

a. -2

b. 3

c. -6

d. 0 ✓

$f'(x) = 3x^2 + 18x = 3(x - (-6))(x - (0))$. Critical points: $x = -6$ and $x = 0$; the larger is 0.

36 [Intermediate] • Target: Rolle's Theorem

Given $-5 \le f'(x) \le 6$ for all x, use the MVT to find the width of the range of possible values for $f(1) - f(0)$.

a. 11 ✓

b. 1

c. 14

d. 12

By the MVT, $f(1) - f(0) = f'(c)(1)$ ranges from $-5(1)$ to $6(1)$, a width of $(6 - (-5))1 = 11(1) = 11$.

37 [Intermediate] • Target: The Mean Value Theorem

Given $-2 \le f'(x) \le 1$ for all x, use the MVT to find the width of the range of possible values for $f(2) - f(-2)$.

a. 3

b. 12 ✓

c. 15

d. 16

By the MVT, $f(2) - f(-2) = f'(c)(4)$ ranges from $-2(4)$ to $1(4)$, a width of $(1 - (-2))4 = 3(4) = 12$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Using the MVT to bound function values • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Does Rolle's Theorem guarantee a $c \in (-4, 6)$ with $f'(c) = 0$ for $f(x) = (x - 1)^2$ on $[-4, 6]$?

a. Yes, but only because f is increasing on the whole interval.

b. Yes — $f(-4) = f(6) = 25$, and f is a polynomial (continuous and differentiable everywhere), so Rolle's Theorem applies; here $c = 1$. ✓

c. No — although $f(-4) = f(6)$, f is not differentiable on $(-4, 6)$, so Rolle's Theorem does not apply.

d. No — Rolle's Theorem never applies to functions of the form $(x - 1)^2$.

Yes — $f(-4) = f(6) = 25$, and f is a polynomial (continuous and differentiable everywhere), so Rolle's Theorem applies; here $c = 1$.

39 [Challenging] • Target: Consequences of the MVT: monotonicity • Cross-Lesson: 8.01 Area Between Curves

Does Rolle's Theorem guarantee a $c \in (-7, 5)$ with $f'(c) = 0$ for $f(x) = (x + 1)^2$ on $[-7, 5]$?

a. Yes, but only because f is increasing on the whole interval.

b. No — Rolle's Theorem never applies to functions of the form $(x + 1)^2$.

c. Yes — $f(-7) = f(5) = 36$, and f is a polynomial (continuous and differentiable everywhere), so Rolle's Theorem applies; here $c = -1$. ✓

d. No — although $f(-7) = f(5)$, f is not differentiable on $(-7, 5)$, so Rolle's Theorem does not apply.

Yes — $f(-7) = f(5) = 36$, and f is a polynomial (continuous and differentiable everywhere), so Rolle's Theorem applies; here $c = -1$.

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40 [Challenging] • Target: Visualizing the Mean Value Theorem • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Does Rolle's Theorem guarantee a $c \in (2, 6)$ with $f'(c) = 0$ for $f(x) = |x - 4|$ on $[2, 6]$?

a. No — Rolle's Theorem never applies to absolute value functions, regardless of the interval.

b. Yes, since $f(2) = f(6)$ is the only hypothesis Rolle's Theorem requires.

c. No — although $f(2) = f(6) = 2$, f has a corner at $x = 4 \in (2, 6)$ and is not differentiable there, so one hypothesis of Rolle's Theorem fails. ✓

d. Yes — $f(2) = f(6) = 2$, and f is continuous and differentiable everywhere, so Rolle's Theorem applies; here $c = 4$.

No — although $f(2) = f(6) = 2$, f has a corner at $x = 4 \in (2, 6)$ and is not differentiable there, so one hypothesis of Rolle's Theorem fails.

41 [Challenging] • Target: Critical Points & First Derivative Test • Cross-Lesson: 1.01 Limits Graphically and Numerically

By the First Derivative Test, what type of extremum does $f(x) = x^3 - 3x^2 - 9x + 5$ have at $x = -1$?

a. Local maximum because f' changes from positive to negative ✓

b. Local minimum because f' changes from negative to positive

c. Inflection point

d. No extremum

For $x < -1$, $f'(x) > 0$, and for $-1 < x < 3$, $f'(x) < 0$. Since f' changes from positive to negative, $x = -1$ is a local maximum.

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42 [Challenging] • Target: Mean Value Theorem (MVT) • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

What are the conditions required to apply the Mean Value Theorem on $[a, b]$?

a. **Continuous on $[a, b]$ and differentiable on (a, b)** ✓

b. Differentiable on $[a, b]$ and $f(a) = f(b)$

c. Continuous on (a, b) only

d. $f'(c) = 0$ on $[a, b]$

The MVT requires that f is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b).

43 [Challenging] • Target: Mean Value Theorem (MVT) • Cross-Lesson: 2.01 The Definition of the Derivative

Why does the MVT NOT apply to $f(x) = |x|$ on $[-2, 2]$?

a. **f is not differentiable at $x = 0$** ✓

b. f is not continuous on $[-2, 2]$

c. $f(-2) \neq f(2)$

d. The interval is not symmetric

$f(x) = |x|$ has a sharp corner at $x = 0 \in (-2, 2)$, violating the differentiability condition.

44 [Challenging] • Target: Rolle's Theorem • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Verify Rolle's Theorem for $f(x) = x^2 - 5x + 4$ on $[1, 4]$. What is the value of c?

a. 2.5 ✓

b. 2.0

c. 3.0

d. 1.5

$f(1) = 0$ and $f(4) = 0$. $f'(x) = 2x - 5 = 0 \Rightarrow c = 2.5 \in (1, 4)$.

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45 [Challenging] • Target: Absolute Extrema on Closed Intervals (Candidates Test) • Cross-Lesson: 3.01 The Chain Rule

Find the absolute maximum value of $f(x) = 2x^3 - 3x^2 - 12x + 1$ on $[-2, 3]$.

a. 8 ✓

b. 1

c. -19

d. -8

Critical points: $f'(x) = 6x^2 - 6x - 12 = 6(x - 2)(x + 1) = 0 \Rightarrow x = -1, 2$. Candidates: $f(-2) = -3, f(-1) = 8, f(2) = -19, f(3) = -8$. Absolute maximum is 8.

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